3.4 Visual Studio and Visual Studio Code

The next two sections deal with two important development environments (*integrated development environments* [IDEs]) from Microsoft: Visual Studio and VS Code. The former can confidently be described as a cornerstone in the history of IDEs. The first version for Windows was launched back in 1997. Until 2014, the proprietary software was only available for a fee; since then, there has been a *Community Edition* that can be used free of charge after registering with Microsoft. The IDE is very popular because it offers context-sensitive help and a syntax check for many supported programming languages. The program requires a lot of resources from your computer, which is why many developers looked for leaner alternatives.

Since 2015, Microsoft has been developing the *VS Code* editor on the *Electron* platform, which was originally created as the basis for the free *Atom* editor. Electron provides a runtime environment for web applications that enables desktop applications without the limitations of the browser sandbox. The core of VS Code is lean and can be extended with countless extensions. In surveys on Stack Overflow, VS Code has been at the top of the list of developers' favorite IDEs for years.

3.4.1 Missing Python Modules in VS Code

The Python application in the following section was already a few years old when we had to deal with it again. An exception occurred when calling the debugger because an integrated module couldn't be found. Unfortunately, the application was created without a Python virtual environment and also without a `requirements.txt` file.

However, the AI support in VS Code based on GitHub Copilot and the corresponding extension were able to save us some work here too. The exception stops the execution and positions the cursor at the corresponding point in the code. The language support for Python in VS Code (again an extension) already indicates the nonexistent import. With Copilot, we can now have a solution suggested at the touch of a button (`/fix`) and insert the corresponding command directly into a console window. Then, simply press the Enter key to install the module.

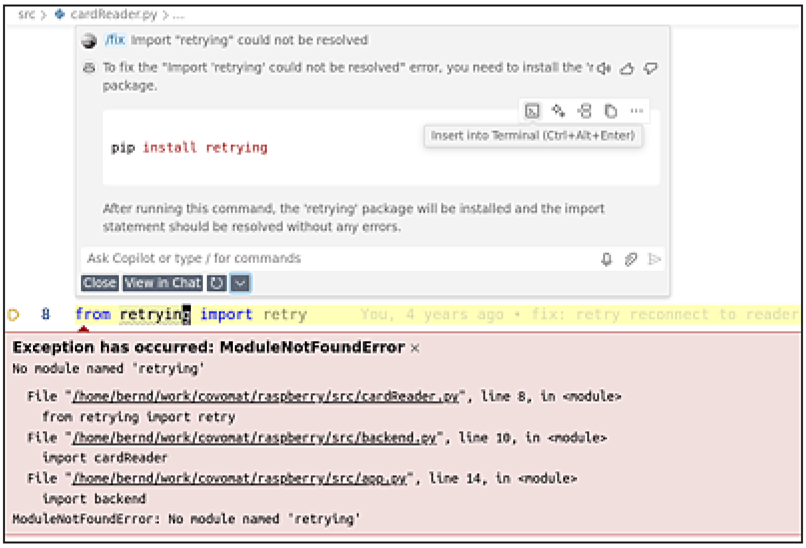

Figure 3.5 GitHub Copilot Can Insert the Call for the Installation of the Missing Python Module Directly into a Terminal

Unfortunately, Copilot in VS Code can't yet handle the text of *exceptions* that occur at runtime. (We'll show how this is possible in Visual Studio in the following section.) However, we suspect that this function isn't too far away.

3.4.2 Visual Studio Exception Debugging

For users of Microsoft's Visual Studio development environment (not the free VS Code), who are also customers of GitHub Copilot, a very convenient AI extension is now available: if you start your application in debugging mode and an unexpected error (an *exception*) occurs, Visual Studio provides the option of having Copilot investigate the problem at the touch of a button.

Our experiments with this function left a very positive impression. The prompt not only sends the text of the exception but also the surrounding code. This was the only way we could understand why the (deliberately caused) `System.ArgumentException` could be explained so precisely. The C# call `todoItems.GetRange(0, 3)` extracts the first three elements from the `todo-`

`Items` list. However, the list was empty when we called it, which is why the exception occurred.

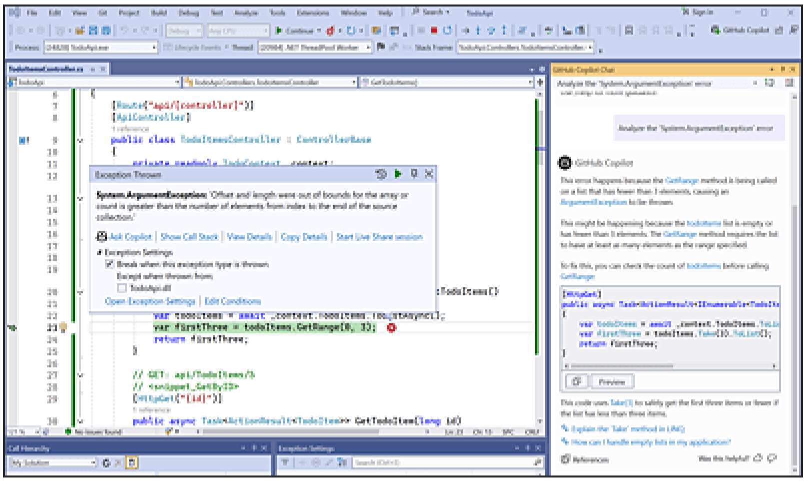

Figure 3.6 GitHub Copilot Support for Debugging in Visual Studio

GitHub Copilot was able to describe the problem briefly and with the correct variables:

> This error happens because the `GetRange` method is being called on a list that has fewer than 3 elements, causing an `ArgumentException` to be thrown.
>
> This might be happening because the `todoItems` list is empty or has fewer than 3 elements. [more details]

The fact that we also got a finished *fix*, that is, a code suggestion that works without any changes, was the icing on the cake. Microsoft is thus demonstrating the convenience of development with AI support. As I said, we're already looking forward to the implementation for VS Code.